CASCADIA ANALYTICS

ASML Holding N.V.

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Exceptional Execution Collides With Valuation Limits

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ASML Holding N.V. continues to demonstrate unprecedented manufacturing scale and operating momentum, underscored by three consecutive guidance raises across fiscal 2026 and record net sales of €11.66B in the third quarter. However, the equity's current market capitalisation of €651.7B and trailing enterprise multiple of 51.2x EBITDA leave virtually no margin for operational friction or shipment normalization. While the company's backlog of €38.80B establishes extraordinary multi-year revenue visibility, the prevailing reference price of $1,675.70 discounts an outcome substantially ahead of intrinsic fundamental value. We initiate coverage with a Hold, Valuation Concerns rating, establishing a probability-weighted blended target of $1,405.23, representing -16.1% upside.

HOLD	$1,405.23	-16.1%	$1,675.70
Hold, Valuation Concerns	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE
INVESTMENT RATING			

The market remains transfixed by quarter-to-quarter shipment cadence and guidance step-ups, failing to account for the valuation multiple compression that historically accompanies terminal transition. While consensus chases the steepening second-half shipment ramp into fiscal 2026, our framework isolates the reality that the business is

already priced for sustained revenue compounding and operating margins well beyond historical norms.

Executive summary

ASML operates as the monopoly supplier of extreme ultraviolet (EUV) lithography systems essential for advanced semiconductor manufacturing, maintaining a critical technological chokepoint across the global semiconductor value chain. The enterprise generates its revenue stream through system sales, comprising EUV, deep ultraviolet (DUV), and metrology solutions, paired with high-margin Installed Base Management covering comprehensive field services and system upgrades. In fiscal 2025, the business generated €32.7B in revenue, delivering a gross profit of €17.3B, an operating income of €11.3B, and free cash flow of €11.0B. This performance translated into an operating margin of 34.6%, a gross margin of 52.8%, and a return on equity of 49.0%, reflecting structural pricing power and unique manufacturing positioning.

The operational trajectory documented across recent quarters highlights robust demand that is actively running ahead of supply capacity. In the third quarter of fiscal 2026, net sales reached a record €11.66B, growing 55.2% year over year and 25% sequentially, supported by a 57.5% gross margin and an Installed Base Management contribution of €3.29B that surpassed guided levels near €2.9B. Management raised full-year 2026 net sales guidance to a range between €44B and €45B, representing the third guidance upgrade of the year, alongside fourth-quarter sales guidance of €14.2B to €15.2B. This followed a second quarter of fiscal 2026 where net sales expanded 21.3% to €9.33B with an operating margin of 37.1%, and an opening first quarter of fiscal 2026 where net sales reached €8.77B with an operating margin of 36.0%. Net bookings in the fourth quarter of fiscal 2025 set a quarterly record at €13.16B, including €7.44B in EUV orders, which elevated the total corporate backlog to €38.80B.

Capital management and liquidity preservation remain pristine, backed by a balance sheet holding €12.9B in cash against total debt of €4.39B, yielding a net cash position of €8.53B at the close of fiscal 2025. This balance sheet strength supported the announcement of a share buyback program of up to €12B alongside the initiation of the original 2026 sales guidance range of €34B to €39B. Free cash flow generation totaled €11.0B in fiscal 2025, representing a 33.8% FCF margin on diluted earnings per share of $24.71. Despite these structural credentials, the equity trades at 67.8x trailing earnings, 19.9x market capitalisation to revenue, and 19.7x enterprise value to revenue. We conclude that current trading levels fully discount operational excellence while leaving the equity exposed to discount rate sensitivity and execution hurdles.

Operational snapshot

$1,675.70	€32.7B	€11.0B	33.8%	1.30	$1,675.70
REF. PRICE	FY2025 REVENUE	FY2025 FREE CASH FLOW	FCF MARGIN	EQUITY BETA	REF. CLOSE

Fiscal-year figures as reported in company filings and collected by Cascadia Analytics; ref. price as of 2026-10-29.

The moat is Absolute and Monopolistic

ASML occupies an unmatched competitive position derived from its status as the exclusive commercial provider of EUV lithography systems. Leading-edge semiconductor fabrication facilities cannot advance node roadmaps without ASML's EUV architecture, which embeds immense switching costs into customer operations. Advanced logic and memory manufacturers design their cleanroom layouts, chemical processes, and transistor interfaces around these specific tools, effectively binding client capital cycles to the company's delivery schedules. These switching barriers are reinforced by high capital commitments, where customer capital expenditure budgets must absorb complex system integration, ensuring exceptional client retention across multi-year fabrication lifecycles.

The regulatory, intellectual, and technical barriers that isolate ASML from potential competitors remain virtually insurmountable. The industrial integration of EUV lithography requires complex light source mechanics, high-precision optical mirrors, vacuum handling, and proprietary metrology interfaces that have taken decades of specialized research to engineer. Beyond system sales, ASML cements its presence through its Installed Base Management business, which generated €3.29B in revenue in the third quarter of fiscal 2026 alone. This division maintains installed equipment through field engineering, proprietary component maintenance, and performance-enhancing upgrades, establishing an ongoing service relationship that generates high recurring cash inflows across economic cycles.

Structural pricing power is evident across the company's reported gross margin profile. ASML achieved a 52.8% gross margin in fiscal 2025 and expanded that metric to 57.5% in the third quarter of fiscal 2026. This margin expansion underscores management's ability to command premium pricing for complex system configurations and service agreements, even as input costs fluctuate across global supply chains. Because customers have no alternative source for EUV tooling, the pricing environment remains firm, allowing ASML to capture a significant share of the total economic value created in advanced semiconductor manufacturing.

Valuation

Three scenarios make the valuation range explicit

Our valuation framework implements an explicit multi-scenario discounted cash flow architecture running from fiscal 2026 through fiscal 2031. Over this forecast window, we calibrate revenue trajectories and free cash flow margins against historical conversion rates and the visible backlog of €38.80B. Cash flows are discounted using a verified weighted average cost of capital of 9.79%, incorporating an equity beta input of 1.30, and capitalization structures reflecting 388.9 million diluted shares. In our base case, we apply a terminal growth rate of 3.0%, reflecting the mature semiconductor equipment replacement cycle.

Under our baseline assumptions, revenue compounds at an 11.5% compound annual growth rate through fiscal 2031, with the target free cash flow margin stabilizing at 33.8% in the final forecast year, matching the 33.8% margin realized in fiscal 2025. This yields a base case equity target of $1,366.55 per share, representing an 18.4% discount to the current reference spot price of $1,675.70. The bull case accelerates revenue expansion to a 14.6% compound annual growth rate, supported by capacity absorption and an elevated final forecast free cash flow margin of 36.0%, deriving an equity value of $1,810.09 per share, which offers an 8.0% premium over current market pricing.

Conversely, our bear scenario captures structural deceleration and cyclical digestion, modeling a 9.0% revenue compound annual growth rate and a compressed final forecast free cash flow margin of 31.4%. This yields a bear target of $1,077.73 per share, representing a 35.7% downside risk from the reference spot price. Applying our institutional probability distribution of 25% bear, 50% base, and 25% bull, we determine a blended target price of $1,405.23 per share, indicating a net negative return expectation of -16.1%. At current trading multiples of 51.2x EV to EBITDA and 19.9x market capitalisation to revenue, the market is pricing expectations beyond our bull case parameters.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	9.0%	31.4%	$1,077.73	-35.7%
Base	50%	11.5%	33.8%	$1,366.55	-18.4%
Bull	25%	14.6%	36.0%	$1,810.09	+8.0%

Bear $1,077.73

Shipment execution slows as client capacity ramps normalize, dragging the revenue compound annual growth rate to 9.0% across

Base $1,366.55

Revenue compounds at an 11.5% annual rate, anchored by the orderly conversion of the €38.80B backlog and steady

Bull $1,810.09

System deliveries and high-margin service contracts accelerate, compounding revenues at 14.6% annually through

the forecast period. Operating friction and lower upgrade utilization press target free cash flow margins down to 31.4%, exposing the stock to a 35.7% decline from the reference spot price.

Installed Base Management growth. Free cash flow margins normalize to the verified fiscal 2025 level of 33.8%, yielding an intrinsic equity value of $1,366.55, or 18.4% below current spot trading.

fiscal 2031. Operating leverage and premium tool pricing expand final forecast free cash flow margins to 36.0%, supporting an intrinsic value of $1,810.09, which sits 8.0% above prevailing spot levels.

Valuation framework

A live discount rate, not a static hurdle

Our quantitative bridge utilizes an institutional capital asset pricing model that establishes a verified discount rate of 9.79%. This cost of capital is parameterized using an equity beta of 1.30, capturing the cyclicality, technological complexity, and macroeconomic exposure inherent in global semiconductor manufacturing equipment. The model accounts for the company's current capital structure, which features 388.9 million diluted shares, net cash of €8.53B, and total debt of €4.39B. This discount rate reflects the risk premium demanded by institutional investors when underwriting hardware tool shipments with multi-quarter lead times.

Discounting cash flows at 9.79% across the fiscal 2026 to fiscal 2031 horizon demonstrates that the stock's current trading valuation incorporates an aggressive duration profile. The market's willingness to price the equity at 67.8x trailing diluted earnings per share implies an artificially suppressed cost of equity or an assumption of perpetual margin expansion. In our discipline, maintaining a rigorous 9.79% discount rate highlights how sensitive terminal asset values are to slight variations in operational delivery, reinforcing our defensive thesis at prevailing spot quotes.

FCF margin normalization is the engine

Cash conversion efficiency represents the critical bridge between reported top-line revenue and shareholder value creation. In fiscal 2025, ASML converted €32.7B of revenue into €11.0B of free cash flow, translating to an FCF margin of 33.8%, supported by €11.3B in operating income and €12.6B in EBITDA. Our valuation scenarios test the resilience of this cash conversion engine under varying operational conditions, shifting from a 31.4% target margin in the bear case to 33.8% in the base case and 36.0% in the bull case.

The progression toward a 36.0% target margin in the bull scenario requires sustained tool delivery schedules and elevated contributions from Installed Base Management, replicating the high-margin environment observed when

gross margin reached 57.5% in the third quarter of fiscal 2026. Conversely, the bear case target margin of 31.4% accounts for potential operational absorption friction, factory floor reinvestment, and fluctuations in working capital. By maintaining 33.8% in our base case, we assume ASML preserves its strong operational efficiency without granting speculative credit for unproven operating leverage.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

> **Methodology.** Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

KEY VALUATION PARAMETERS

Forecast window	**FY2026 to FY2031**
Live WACC	**9.79% (2026-10-29)**
Terminal growth	**3.0%**
Equity beta	**1.30**
Diluted shares	**388.9 million**

HOLD

over $1,405.23 blended target

Competitive moat and concentration risk

We rate the moat as Wide and we view the market structure as Monopoly (EUV lithography). We assign a premium of +35% to sector median multiples on moat strength alone.

- sole-supplier EUV + High NA, first High NA shipments 2023, Intel past 1m High NA wafers, Samsung and SK Hynix adopting High NA from 2028 (Reuters, Sep 2026).
- FY2025 net sales €32.7bn (+15% YoY), EPS €24.73 (+28.5%), 2026 guidance €34-39bn at 51-53% gross margin, year-end 2025 backlog €38.8bn (company/press coverage, 2026); SK Hynix disclosed a ~$8bn EUV order on 24 Mar (delivery by 31 Dec 2027, ~30 tools at $250-300m each, Bernstein estimate). A €38.8bn backlog at 51-53% guided gross margin is durable pricing power, not a spot spike.
- The weakest element is geopolitical/regulatory exposure, export controls on China sales and single-customer concentration (TSMC, Samsung, Intel, SK Hynix, Micron) cap the rating below an unconditional premium, but do not remove Wide, no rival can replicate the installed base, service network, or Zeiss optics chain within a planning horizon.

On a multiple cross-check, the justified range is 24.0-27.0x / 20.9-23.5x. The cross-check value is $1,313-$1,477 vs Base $1,203.94, spot $1,603.91, on a base of Framework base from the October 27 moat report. The gap verdict is In part, materially (~68% of gap at top); spot P/E 25.5x still above justified ceiling.

Backlog is €38.8bn system backlog, 31 Dec 2025, with coverage of 14.2 months vs FY2025 sales, 10.5 months vs 2026 guide and visibility rated High. On backlog alone, the stance is Supports.

We score concentration and credit risk across seven families, where a High score is 8 points, a Medium score is 4 points, and a Low score is 1 point.

Risk family	Score
customer concentration	8 points
counterparty credit and funding dependence	1 points
single product, single program, or single platform dependency	4 points
single supplier, single fab, or single geography exposure	8 points
platform, channel, or distribution dependency	1 points
customer insourcing risk	1 points
regulatory single point risk	8 points

The combined discount is 20 percent, capped at 20 percent, against an uncapped total of 31 percent. Shown side by side with the moat premium or discount, the net stance is +15 pp points arithmetically. We do not net these figures into a new target, and we do not double count a family excluded under the double-count rule.

The ladder ranks risk coarsely, and because 32 of the 47 names reach the 20 percent cap, the combined discount is not a mechanical valuation haircut until the ladder is recalibrated.

On the gap to spot, the concentration assessment reads as follows. Not at all

Both the moat premium or discount and the concentration discount are multiple and risk cross-checks dated October 27, 2026, and neither figure is in the DCF target itself. Cross-check figures were computed against the Base in the October 27 framework, so they may differ from a live Base that has since moved.

Business & segment drivers

Revenue prospects anchor on Extreme Ultraviolet Tool Deliveries and Installed Base Management Services HOLD

ASML's top-line performance rests on two foundational growth engines, namely system sales comprising EUV, DUV, and metrology platforms, and high-margin Installed Base Management. Extreme ultraviolet lithography systems represent the company's primary technological franchise, enabling advanced node production across leading foundries and integrated device manufacturers. System shipments require extensive planning cycles, as evidenced by the fourth quarter of fiscal 2025, where net bookings reached a record €13.16B with EUV orders accounting for €7.44B. This sustained intake expanded the order backlog to €38.80B, giving ASML exceptional revenue coverage as shipments ramp across subsequent fiscal cycles.

Alongside new tool shipments, the Installed Base Management division provides an indispensable recurring revenue layer. In the third quarter of fiscal 2026, Installed Base Management delivered net sales of €3.29B, surpassing internal guidance that centered near €2.9B. This business encompasses ongoing maintenance, field modifications, and system productivity upgrades designed to maximize wafer output across mature and advanced tool installations. Because this service business scales in lockstep with the cumulative volume of machines operating worldwide, it generates consistent high-margin cash flow that partially insulates corporate profitability from hardware delivery timing.

The operating trajectory observed across fiscal 2026 demonstrates this dual-engine dynamic in practice. Net sales expanded from €8.77B in the first quarter of fiscal 2026 to €9.33B in the second quarter, before surging to a record €11.66B in the third quarter of fiscal 2026. Operating margins advanced simultaneously, moving from 36.0% in the first quarter to 37.1% in the second quarter, while gross margin expanded to 57.5% in the third quarter. These operational milestones supported three successive upward revisions to full-year 2026 sales guidance, which moved from the initial range of €34B to €39B announced in late fiscal 2025 to a finalized expectation of €44B to €45B by the third quarter of fiscal 2026.

What to monitor

- **Guidance execution across the fourth quarter of fiscal 2026** Tracking reported net sales against management's guided window of €14.2B to €15.2B to verify shipment throughput.
- **Installed Base Management run rates** Monitoring whether quarterly service revenue sustains levels above the €3.29B milestone achieved in the third quarter of fiscal 2026.
- **Backlog replenishment and order conversion** Evaluating order intake relative to the €38.80B backlog benchmark established in the fourth quarter of fiscal 2025.
- **Operating profitability sustainability** Assessing operating margin performance against the 37.1% level recorded in the second quarter of fiscal 2026 and the 57.5% gross margin delivered in the third quarter.

+15.6%	52.8%	34.6%	33.8%	€8.53B	49.0%
FY2025 REVENUE GROWTH	GROSS MARGIN	OPERATING MARGIN	FCF MARGIN	NET CASH	RETURN ON EQUITY

METRIC	FY2025 (AS REPORTED)
Revenue	€32.7B
Gross Profit	€17.3B
EBIT	€11.3B
Net Income	€9.61B
EBITDA	€12.6B
Free Cash Flow	€11.0B
Diluted EPS	$24.71
ROE	49.0%

Risk & moat matrix

The bear case

SHARE EROSION

While ASML holds a monopoly in commercial EUV lithography systems, customer concentration across leading foundries creates single-point dependency on capital expenditure cycles. Any adjustment in customer node transitions directly threatens

M&A DIGESTION

Balance sheet gains such as the Mistral AI investment revaluation recorded in the third quarter of fiscal 2026 can obscure underlying operational trends. Over-reliance on ancillary equity appreciation introduces potential earnings volatility that

REGULATORY AND PRICING SCRUTINY

Global semiconductor trade restrictions and cross-border equipment controls remain an ongoing risk for complex tool sales across EUV and DUV lines. Changes in export compliance frameworks could impede delivery timelines and disrupt

TECHNOLOGY DISRUPTION

The manufacturing transition toward next-generation lithography tools requires flawless execution across complex optical and vacuum architectures. Engineering bottlenecks or component delays could trigger delivery postponements and alter customer deployment schedules.

REINVESTMENT TREADMILL

Maintaining technological leadership across EUV, DUV, and metrology necessitates persistent research and development allocation alongside heavy capital expenditure. Elevated ongoing capital needs could restrict free cash flow conversion toward the lower bounds of our scenario model.

WACC AND DURATION SENSITIVITY

Trading at 67.8x trailing earnings and an enterprise value to revenue multiple of 19.7x leaves the equity vulnerable to interest rate movements. Under a 9.79% cost of capital, any delay in terminal cash generation triggers significant downward re-rating in market value.

shipment absorption across system sales.	distracts from pure-play equipment delivery execution.	backlog fulfillment.			

The strengths

OLIGOPOLY POSITION

ASML stands as the sole global provider of EUV lithography systems, giving the enterprise an undisputed commercial monopoly over advanced node manufacturing equipment. This structural position creates pricing power that supported a 57.5% gross margin in the third quarter of fiscal 2026.

FCF COMPOUNDING

The business converted 33.8% of fiscal 2025 revenue directly into €11.0B of free cash flow, anchored by €12.6B in EBITDA and €11.3B in operating income. This high cash conversion provides balance sheet flexibility and funded the initiation of an up to €12B share buyback program.

GEOGRAPHIC DURABILITY

A corporate backlog of €38.80B, established with record net bookings of €13.16B in the fourth quarter of fiscal 2025, spans leading semiconductor manufacturing regions worldwide. This extensive order book guarantees multi-year factory floor utilization and predictable tool shipment schedules.

PIPELINE OPTIONALITY

Installed Base Management generated €3.29B in net sales during the third quarter of fiscal 2026, providing high-margin service and upgrade revenue. As the operational footprint of installed EUV and DUV systems broadens, recurring service agreements compound independent of new tool shipments.

The bottom line on risk and reward

The investment narrative surrounding ASML presents an unmistakable divergence between fundamental operational execution and equity valuation discipline. Operationally, the company demonstrates unparalleled strength, having raised full-year 2026 net sales guidance three consecutive times to reach €44B to €45B, backed by an implied fourth-quarter step-up to between €14.2B and €15.2B. Furthermore, the generation of €11.0B in free cash flow on €32.7B of revenue in fiscal 2025 confirms the cash-generative power of its lithography monopoly. These attributes place ASML among the most strategically vital technology enterprises globally.

However, capital allocation requires weighing operational quality against prevailing asset prices. At the reference spot price of $1,675.70, ASML trades at 67.8x trailing earnings, 19.9x revenue, and 51.2x EBITDA, representing market pricing that already assumes flawless execution through the terminal forecast. Our probability-weighted discounted cash flow model yields a blended

price target of $1,405.23, which implies -16.1% upside. Even our optimistic bull case target of $1,810.09 offers only an 8.0% return, while the bear case target of $1,077.73 warns of -35.7% downside risk.

Because the risk-to-reward ratio tilts unfavorably, we see no quantitative justification for initiating or expanding long positions at current trading multiples. Market expectations demand continual guidance upgrades and unblemished shipment delivery, leaving the share price highly sensitive to minor earnings disruptions or discount rate recalibrations. We reiterate our Hold, Valuation Concerns rating, advising institutional clients to await a valuation contraction toward our base case intrinsic level of $1,366.55 before committing incremental capital.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

Franchise deep dive

EUV Lithography Systems: The Monopoly Chokepoint of Advanced Semiconductor Manufacturing

ASML's extreme ultraviolet (EUV) lithography systems franchise forms the central pillar of advanced semiconductor manufacturing globally. Operating as the sole commercial supplier of EUV equipment, the company provides the essential photolithography systems required by leading fabricators to print microscopic circuit structures on silicon wafers. Without these advanced EUV tools, semiconductor fabricators cannot manufacture cutting-edge processors, solidifying ASML's status as an indispensable supplier across the global digital economy.

The commercial strength of this franchise is illustrated by its order book velocity and backlog composition. In the fourth quarter of fiscal 2025, ASML recorded total net bookings of €13.16B, of which €7.44B represented EUV systems alone, lifting the total corporate backlog to €38.80B. This equipment backlog provides multi-year revenue visibility, allowing management to optimize capital outlays and factory floor throughput across production facilities. Complementing the hardware platforms, ASML's metrology and deep ultraviolet (DUV) systems work alongside the EUV fleet to provide complete lithography solutions across advanced and mature semiconductor nodes.

Complementing new tool shipments is the Installed Base Management division, which extracts ongoing economic value from every deployed system. In the third quarter of fiscal 2026, Installed Base Management generated net sales of €3.29B, exceeding internal projections near €2.9B. This business encompasses continuous maintenance services, hardware upgrades, and productivity enhancements that extend system longevity. The compounding nature of the installed tool fleet provides ASML with a high-margin recurring revenue stream that supports gross margins, which reached 57.5% in the third quarter of fiscal 2026.

What the recent record shows

- Record quarterly net bookings of €13.16B achieved in the fourth quarter of fiscal 2025, featuring €7.44B in dedicated EUV orders that elevated total corporate backlog to €38.80B.
- Third-quarter fiscal 2026 net sales reached a record €11.66B, advancing 55.2% year over year and supported by an Installed Base Management revenue contribution of €3.29B.
- Management raised full-year 2026 sales guidance for the third consecutive time to €44B to €45B, accompanied by fourth-quarter sales guidance of €14.2B to €15.2B.
- Gross margin reached 57.5% in the third quarter of fiscal 2026, expanding from 52.8% in fiscal 2025, illustrating the pricing power inherent in the EUV system and service ecosystem.

Our read: ASML's operational monopoly across EUV lithography systems and Installed Base Management services remains completely uncontested, generating record sales and cash flows. Nevertheless, with the equity trading at 51.2x EBITDA and offering a blended target of $1,405.23, representing -16.1% upside against the reference spot price, the stock fully discounts this operational brilliance, maintaining our Hold, Valuation Concerns stance.